Fundamentals of Machine Learning
Mock Midterm: Solutions

Every part answered in the shape the grader wants: claim, mechanism, number or derivation, consequence

Fundamentals of Machine Learning

2026-10-30

1. Account takeover

(a) Object: one login attempt. x: the five recorded features at the moment of the login. $Y \in \{\text{legitimate}, \text{takeover}\}$, assigned later from disputes. Model output: $\hat{p}(y{=}1 \mid x) \in [0,1]$. Actions: $\{\text{allow}, \text{block}, \text{code}\}$. Learning signal: the labelled pairs (x_i, y_i), available only 60 days after each login.

(b) "Support ticket within 7 days" is *leakage*: it happens after the login, is a consequence of the takeover being discovered, and is not available at decision time. "Login at night" is legitimate information (known at decision time) but a *potential shortcut*: if the collected takeovers happen to cluster at night, the model learns the hour rather than the behaviour, and fails when attackers change schedule. Leakage is removed by construction; a shortcut is only detected by data that breaks the correlation (daytime takeovers, night-time legitimate logins).

(c) Risks: $R(\text{allow}) = 30p$; $R(\text{block}) = 1 \cdot (1-p)$; $R(\text{code}) = 0.5 + (1-q) \cdot 30\,p = 0.5 + 3p$. Allow beats code when $30p < 0.5 + 3p$, $p < 0.5/27 = 0.0185$, and it then also beats block ($30p < 1 - p$ for $p < 0.032$). Block beats code when $1 - p < 0.5 + 3p$, $p > 0.125$. Regions: allow for $p < 0.0185$, code for $0.0185 \le p \le 0.125$, block for $p > 0.125$. The imperfect code tilts its risk line upward (slope 3), which shrinks the middle band compared with a perfect review.

(d) $r_d = \dfrac{(\pi_d/\pi_t) r_t}{(\pi_d/\pi_t) r_t + \frac{1-\pi_d}{1-\pi_t}(1 - r_t)} = \dfrac{0.006 \times 0.8}{0.0048 + 1.994 \times 0.2} = \dfrac{0.0048}{0.4036} = 0.0119$. Since $0.0119 < 0.0185$, the action is *allow*. Without the correction the system would read 0.80 and block; with it the login goes through. With three actions there are two thresholds and both sit below 0.13, so an uncorrected posterior inflated by a factor near 160 in the odds lands every login in the block region: the correction is not a refinement, it decides everything.

(e) A dispute is filed only if the customer notices; unnoticed takeovers are labelled legitimate. The features can be perfect and the labels still noisy, which raises the Bayes error: no model on this x can be right on the mislabelled fraction.

2. The loss decides

(a) Squared loss: $a = \mathbb{E}[Y] = 0.4 + 0.6 + 1.0 + 2.0 = 4.0$ days. Absolute loss: the median, the smallest a with $F(a) \ge 0.5$: $F(1) = 0.4$, $F(2) = 0.7$, so $a = 2$. The mean 4.0 is never observed; that does not matter for squared loss, which minimises the average squared distance, not the probability of being exactly right. It matters if the number is read as "the typical case".

(b) With $c_u = 9$ (under) and $c_o = 1$ (over), the optimum is any τ-quantile with $\tau = c_u/(c_u + c_o) = 0.9$: the set of a with $\mathbb{P}(Y < a) \le 0.9 \le \mathbb{P}(Y \le a)$. $F(5) = 0.9$ exactly, so every $a \in [5, 20]$ satisfies it: the expected loss is flat on that interval. Check: the derivative of the risk on $(5, 20)$ is $c_o \mathbb{P}(Y \le a) - c_u \mathbb{P}(Y > a) = 0.9 - 0.9 = 0$.

(c) $a = 2$: $0.4 \times 1 + 0 + 0.2 \times 27 + 0.1 \times 162 = 22.0$. $a = 5$: $0.4 \times 4 + 0.3 \times 3 + 0 + 0.1 \times 135 = 16.0$.

$a = 20$: $0.4\times19+0.3\times18+0.2\times15 = 16.0$. Predicting 20 costs the same as predicting 5 but is not safer: it pays 15 days of over-prediction on 90% of the hosts to save 15 days of under-prediction on 10%; the loss already prices caution at 9:1 and the quantile is exactly where the two sides balance.
(d) The model would need the conditional distribution, or at least its 0.9-quantile. Options besides retraining: estimate the conditional distribution once (quantile regression at several levels, or a probabilistic model) and derive any loss's optimum from it; the mean alone cannot be converted into a quantile.

3. Capacity, bias and variance, protocol

(a) Over repeated draws of the responses at the fixed inputs, $\text{bias}(0) = \mathbb{E}[\hat{f}(0)] - f(0)$ and $\text{Var}(\hat{f}(0)) = \mathbb{E}[(\hat{f}(0) - \mathbb{E}\hat{f}(0))^2]$; the expected squared error is $\sigma^2 + \text{bias}^2 + \text{Var}$ with $\sigma^2 = 1$.
(b) $\hat{f}(0)$ averages k responses $f(x_j) + \varepsilon_j$, so bias $= \frac{1}{k}\sum f(x_j)$ and variance $= 1/k$. $k = 1$: bias 0, variance 1, error $1 + 0 + 1 = 2.00$. $k = 3$: bias $(0 + 0.25 + 0.25)/3 = 0.167$, variance 0.333, error $1 + 0.028 + 0.333 = 1.36$. $k = 5$: bias $(0 + 0.25 + 0.25 + 1 + 1)/5 = 0.5$, variance 0.2, error $1 + 0.25 + 0.2 = 1.45$. Best $k = 3$. Raising k buys variance $(1/k)$ with bias (farther neighbours, whose f differs from $f(0)$).
(c) Denser data puts the k nearest points closer to 0, so the bias term shrinks for every k while the variance $1/k$ is unchanged; larger k becomes affordable and the best k grows. Capacity is relative to sample size.
(d) (1) Scaler fitted on all rows: test statistics shape training inputs; inflates every detector slightly. (2) Feature selection on all rows, including the validation folds: the 30 features were chosen with knowledge of the labels the folds will test; inflates every detector, strongly when d is large. (3) Selecting the best of four on the same folds and reporting that score: the winner's curse; inflates only the selected one. (4) If attacks produce near-duplicate rows and folds are random, the training folds contain the test folds; inflates every detector.
(e) Split by attack episode (and by time if drift is expected) into development and test. Inside the development set run K-fold CV where each fold fits the scaler, the feature selection and the detector on its training part only. Pick the detector and its configuration by the CV score. Refit on the whole development set. Score the test set once. Report that, with precision, recall and the cost-weighted error at the chosen threshold.
(f) $\mathbb{E}[\min] = R - \sigma/\sqrt{\pi} = 0.10 - 0.015/1.772 = 0.10 - 0.0085 = 0.0915$. With 50 detectors the expected minimum of 50 noisy draws is lower still (the optimism grows roughly like $\sigma\sqrt{2\ln 50}$). With a larger validation set σ falls, so the optimism shrinks: the selection bias is noise times the number of things selected among.

4. Generative models

(a) $\log\frac{p(1|x)}{p(0|x)} = \log\frac{\pi_1}{\pi_0} + \log p(x \mid 1) - \log p(x \mid 0) = \log\frac{0.1}{0.9} + \left[-\frac{1}{2}\log(2\pi\cdot 4) - \frac{(x-2)^2}{8}\right] - \left[-\frac{1}{2}\log(2\pi) - \frac{x^2}{2}\right] = -\ln 9 - \ln 2 + \frac{x^2}{2} - \frac{(x-2)^2}{8}$. The x^2 terms do not cancel because the variances differ ($\frac{1}{2} - \frac{1}{8} = \frac{3}{8} \neq 0$); with equal variances they would, leaving a linear function (LDA).
(b) Multiply by 8: $4x^2 - (x^2 - 4x + 4) = 8\ln 18$, so $3x^2 + 4x - 4 - 23.12 = 0$, $x = \frac{-4\pm\sqrt{16+325.4}}{6}$: $x = 2.41$ and $x = -3.75$. Attack region: $x > 2.41$ or $x < -3.75$. Two boundaries, because the wider class-1 density overtakes the narrow class-0 density on both sides.
(c) At $x = 10$: $-2.89 + 50 - 8 = 39.1$, posterior $\sigma(39.1) \approx 1$. Both likelihoods are astronomically small (e^{-50} and e^{-8}); the model is confident only about the *ratio*: a point at 10 is far less implausible under the wide Gaussian than under the narrow one. A generative model can also report $p(x) = \sum_c \pi_c p(x \mid c)$, which is near zero here: the analyst wants "this flow is unlike anything seen", a reject or novelty flag, not "attack with probability 1".
(d) Equal variances: the quadratic terms cancel, the log-odds is linear in x, and there is a single

boundary point; the posterior still goes to 0 or 1 far away, but monotonically on one side.
(e) Odds $= 1 \times 3 \times 3 = 9$, posterior 0.9. With x_2 a copy of x_1 the evidence is one feature: true odds 3, posterior 0.75; naive Bayes multiplies the duplicated ratio and reports 0.9. The reported score is a monotone function of the true one (here odds^2), so the ordering of flows is preserved and the threshold can be re-tuned; the calibration is wrong, which breaks any cost-based decision that trusts the number.
(f) Prefer logistic regression because the two features are dependent and it learns one weight for the combined evidence instead of double counting. Prefer the generative model when attacks are a few dozen examples: it estimates two Bernoulli parameters per class from each feature's own counts, is data efficient, and can handle a missing feature by dropping its factor.

5. Ensembles

(a) $\mathrm{Var}(\bar{f}) = \frac{1}{M^2}\left[M\sigma^2 + M(M-1)\rho\sigma^2\right] = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{M} = 0.4 + 0.6/M$. At most 0.45: $0.6/M \le 0.05$, $M \ge 12$. The floor as $M \to \infty$ is $\rho\sigma^2 = 0.4$: averaging removes only the uncorrelated part.
(b) Decorrelated: $0.24 + 0.96/M$. Equal when $0.16 = 0.36/M$, $M = 2.25$; for $M \ge 3$ the decorrelated forest is better, and its floor is 0.24 against 0.4. Each tree is worse (variance 1.2), the forest is better because the floor, not the single tree, decides: weaker trees, stronger forest.
(c) $g_i = p_i - y_i = (-0.7, 0.6)$, $h_i = p_i(1-p_i) = (0.21, 0.24)$, $G = -0.1$, $H = 0.45$. Leaf value $w^* = -G/(H+\lambda)$: 0.222 for $\lambda = 0$, 0.069 for $\lambda = 1$. The gradients have opposite signs: the positive example wants the logit up, the negative wants it down, and they nearly cancel, so the leaf moves both by a small amount in the wrong direction for one of them. The tree should split them apart so each gets its own correction; a leaf that mixes conflicting gradients is the signature of an under-grown tree.
(d) Singletons: $\frac{g_L^2}{h_L+1} = \frac{0.49}{1.21} = 0.405$, $\frac{g_R^2}{h_R+1} = \frac{0.36}{1.24} = 0.290$, parent $\frac{0.01}{1.45} = 0.007$. Gain $= \frac{1}{2}(0.405 + 0.290 - 0.007) - \gamma = 0.344 - \gamma$. Kept while $\gamma < 0.344$.
(e) In a forest each tree never saw its out-of-bag rows, so those rows are a test set for that tree. A boosted model is one sequence of trees each fitted to the residuals of the previous ones on the *same* rows; no row is unseen by the ensemble, so evaluation needs a held-out set or cross-validation.

6. Four claims

(a) Wrong. With 0.3% takeovers the rule "always allow" has accuracy 99.7%, higher than the detector; accuracy is dominated by the majority class. Report recall and precision on the takeovers and the cost-weighted error at the operating threshold.
(b) Wrong. The 20% was used to select among 50 configurations, so it is a validation set and 0.93 is the maximum of 50 noisy scores: the winner's curse (Problem 3(f) in miniature). Test performance needs a set touched once, after selection.
(c) Right only under a condition. Averaging bounds the variance from below at $\rho\sigma^2$ and cannot remove bias or a shortcut; a forest of deep trees on leaked or duplicated rows overfits in the only sense that matters, the gap to deployment. It is robust to overfitting *from adding more trees*, which is the narrow claim that is true.
(d) Wrong. The training posterior is calibrated to a 50% prior; at deployment with 0.3% takeovers it overstates the odds by a factor $\frac{0.5/0.5}{0.003/0.997} \approx 330$. Either correct it by prior shift and then apply the cost threshold $C_{FP}/(C_{FP}+C_{FN})$, or choose the threshold on deployment-distribution validation data; 0.5 is right only if the deployment prior is also one half.

If you missed a problem, reread: 1: Essential ch. 1 §3, §5, ch. 2 §2, ch. 4 §6. 2: ch. 2 §3 and derivation 2 of *How the Professor Asks*. 3: ch. 3 §3, §5–6. 4: ch. 4 §3, §4, §6. 5: ch. 5 §1, §3, §6. 6: the sixteen pairs in *How the Professor Asks*, section 2.